Prodigy Anesthesia Quick Review

Cor Pulmonale

What is cor pulmonale?	Cor pulmonale is characterized by pulmonary hypertension that progresses to right ventricular hypertrophy, dilation, and eventual cardiac failure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 627-628.*
What are the treatment options for cor pulmonale?	Hypoxemia worsens pulmonary hypertension and should be treated both acutely and chronically by the administration of oxygen. Diuretics and digitalis may be used in episodes of right ventricular heart failure that don't respond to correction of arterial blood gases. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 141-142.*
What factors determine the prognosis of cor pulmonale?	Patients who can maintain a normal PAO2 have a favorable prognosis. Patients with concomitant hypoxic lung disease have a 70% mortality rate within five years from the onset of peripheral edema. Patients who develop cor pulmonale as a result of an intrinsic pulmonary vascular disease such as pulmonary fibrosis have an extremely poor prognosis. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 628.*
How does the incidence of cor pulmonale in males compare to that in females?	Cor pulmonale is five times more likely to occur in male patients *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 628.*
What are the primary anesthetic management objectives for the patient with cor pulmonale?	Preoperative goals are centered around hydration, correction of electrolyte abnormalities, treating bronchospasm, and improving the clearance of pulmonary secretions. Large doses of opioids should be avoided intraoperatively as they may increase the risk of postoperative ventilatory depression. Volatile anesthetics assist in bronchodilation and positive-pressure ventilation aids in correcting V/Q mismatching and opening collapsed alveoli. Regional anesthesia is effective for lower extremity procedures, but should be avoided if a high level is required that would impair the accessory muscles of respiration. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 141-142.*
How does chronic hypoxemia and hypercarbia result in cor pulmonale?	Chronic hypoxemia and hypercarbia result in constriction of pulmonary arterioles and venules. The vasoconstriction in pulmonary vessels results in pulmonary hypertension. The pressure in the pulmonary circuit is transmitted to the right ventricle, resulting in right-sided heart failure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 141.*

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What radiographic signs may be seen in patients with cor pulmonale?	The chest xray often reveals increased width of the right pulmonary artery and decreased pulmonary vascular markings in the lung periphery. A late sign of cor pulmonale may be a decrease in the retrosternal space on the lateral film as right ventricular enlargement becomes evident. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 141-142.*
What ECG changes may be seen in patients with cor pulmonale?	Right atrial hypertrophy is manifested as peak P waves in leads II, III, and aVF. Right axis deviation and right bundle branch block are often present as a result of right ventricular hypertrophy. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 141-142.*

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